F79SP Session 3: The Ruin Problem

1 The average process

The **average process** of a random walk $\{S_n\}_{n\geq 0}$ is $\{\frac{S_n}{n}\}_{n\geq 1}$. For the case when $S_0 = 0$, we can think of each $\frac{S_n}{n}$ as the average of increments.

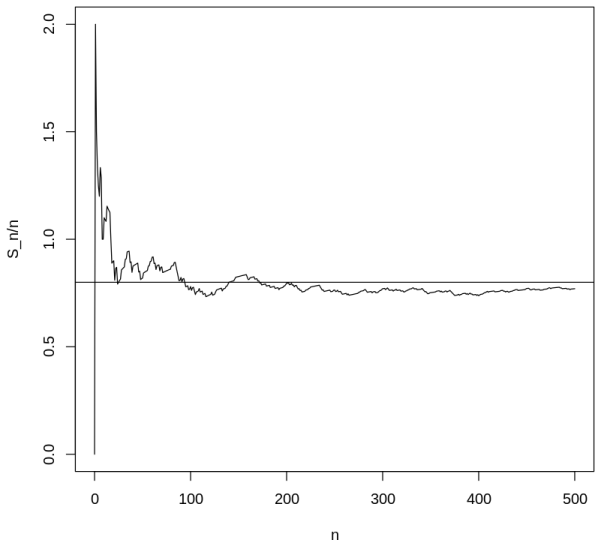


Figure 1: A realisation of average process of a modification of random walk in Example 2.1., Session 2, with $a = 0$. The random walk has $\mu = \mathbf{E}\xi_1 = 0.8 > 0$. Observe the difference in behaviour between the RW and its average process.

General discussion

Assume, for simplicity, that $a = 0$, i.e. $S_n = \sum_{i=1}^{n} \xi_i$. Note that

$$\mathbf{E}\left(\frac{S_n}{n}\right) = \frac{n\mu}{n} = \mu \qquad \text{for all } n$$
$$\mathbf{Var}\left(\frac{S_n}{n}\right) = \frac{\mathbf{Var}S_n}{n^2} = \frac{n\sigma^2}{n^2} = \frac{\sigma^2}{n} \qquad \to 0 \quad \text{as } n \to \infty.$$

We also have, from Chebyshev's Inequality, that, for any $a > 0$,

$$\mathbf{P}\left(\left|\frac{S_n}{n} - \mu\right| \geq a\right) \leq \frac{\mathbf{Var}(S_n/n)}{a^2}$$
$$= \frac{\sigma^2}{na^2} \qquad \to 0 \quad \text{as } n \to \infty,$$

i.e. as $n \to \infty$, the process $\{S_n/n\}_{n\geq 1}$ tends to cluster ever closer around the constant μ.

Indeed the Strong Law of Large Numbers (SLLN) states that

$$\frac{S_n}{n} = \frac{1}{n}\sum_{i=1}^{n} \xi_i \to \mu \quad \text{almost surely, as} n \to \infty, \quad \text{that is,} \mathbf{P}\left(\frac{S_n}{n} \to \mu\right) = 1.$$

2 The ruin problem

Starting at $a > 0$, does the random walk reach $S_n = 0$ or below for some $n > 0$?

This event is defined as ruin, denoted as R. It is also good to gain insights on when ruins are expected to **first** occur? Define:

$$T_R = \begin{cases} \text{First } n > 0 \text{ such that } S_n \leq 0 \\ \infty, \text{ if } S_n > 0 \text{ for all } n \end{cases}$$

Three points of interest:

- The ruin probability: $\mathbf{P}(R) = \mathbf{P}(\text{there exists } n > 0 \text{ such that } S_n \leq 0)$.
- Since T_R is a r.v., we are interested with the expected time to ruin: $\mathbf{E}T_R$.
- The probability that ruin occurs by some fixed time N: $\mathbf{P}(T_R \leq N)$. Challenging!

A classification is provided based on $\mu = \mathbf{E}\xi_i$.

Case 1: $\mu < 0$ (process drift downwards)

We have $\mathbf{P}(R) = 1$ and $\mathbf{E}T_R < \infty$.

1. $\mathbf{P}(R) = 1$ follows from SLLN. As $\mathbf{P}(\frac{S_n - a}{n} \to \mu) = 1$, gives $S_n \to n\mu + a$ almost surely. Sufficiently large n will guarantee $n\mu + a < 0$ since $\mu < 0$ and a is fixed. Hence, there exists $n > 0$ such that $S_n < 0$.
2. $\mathbf{E}T_R < \infty$ and intuitively for large a, $\mathbf{E}T_R \approx -\frac{a}{\mu}$.
3. For N large, CLT can be used to estimate $\mathbf{P}(T_R \leq N)$.

For large N:

$$\begin{aligned}\mathbf{P}(T_R \le N) &= \mathbf{P}(T_R < N \cup S_N \le 0) \\ &\ge \mathbf{P}(S_N \le 0) \\ &= \mathbf{P}\left(\frac{S_N - (a+N\mu)}{\sigma\sqrt{N}} \le -\frac{a+N\mu}{\sigma\sqrt{N}}\right) \\ &\approx \Phi\left(-\frac{a+N\mu}{\sigma\sqrt{N}}\right)\end{aligned}$$

Taking $N \to \infty$, we can see that $\Phi\left(-\frac{a+N\mu}{\sigma\sqrt{N}}\right) \to 1$ if N is relative large to a, so $\mathbf{P}(T_R \le N)$ would be close to 1.

Case 2: $\mu = 0$ (process with zero drift)

We have $\mathbf{P}(R) = 1$ and $\mathbf{E}T_R = \infty$.

- Comment: $\mathbf{P}(R) = 1 = \sum_{n=1}^{\infty} \mathbf{P}(T_R = n)$ but $\mathbf{E}T_R = \sum_{n=1}^{\infty} n\mathbf{P}(T_R = n) = \infty$.

Case 3: $\mu > 0$ (process drift upwards)

We have $\mathbf{P}(R) < 1$ and $\mathbf{E}T_R = \infty$.

1. $\mathbf{P}(R) < 1$. For most distribution of increment ξ, $\mathbf{P}(R) < e^{-\gamma a}$ for some adjustment coefficient $\gamma > 0$ (Lundberg's inequality)
2. $\mathbf{E}T_R = \infty$ because $\mathbf{P}(T_R = \infty) > 0$.